Atomquest 2025 - Robotic Cleaning Bot Specification Sheet

General Information

Parameter	Specification
1. Team Name	Hack O' Giants
2. Institute Name	Progressive Education Society's Modern College of Engineering, Pune
3. Team Members	Gurjas Singh Gandhi (SY MCA) Joy Johnson Kujur (SY MCA) Sumedh Khedekar (SY MCA) Prathamesh Lohar (SY MCA) Dipali Gode (SY MCA) Swapnil Kulkarni (FE Mechanical)
4. Bot Name	CHOTU – "Compact Home Operations & Tidying Unit"
5. Control Type	Autonomous

Physical Specifications

Parameter	Specification
1. **Dimensions**	1.5 ft x 1.5 ft x 1 ft
2. **Weight**	2.4 kg
3. **Chassis Material**	MDF Board — for the sturdy prototype body. *"3D Printed PLA parts — for precise, lightweight framing."*
4. **Wheels/Drive**	2 motorized wheels + 1 caster wheels

Power System

Parameter	Specification
1. **Battery Capacity**	11.1 V, 2 Ah , 3000mh 11.1 V, 2 Ah , 6000mh
2. **Battery Life (Single Charge)**	1 hour 24 minutes
3. **Charging Time**	~3.3–3.6 hours
4. **Power Consumption**	**≈ 47.7 W**

Motors

Parameter	Specification
1. **Drive Motors**	2 x BO motor 200rpm
2. **Suction Motor**	12 V DC motor
3. **Brush Motor**	2 x BO motor 200rpm

Microcontroller

Parameter	Specification
Microcontroller	Seed Studio XIAO ESP32S3 Sense CPU: Dual-core, 240 MHz Connectivity: Wi-Fi, Bluetooth Integrated Sensors: OV2640 Camera Sensor, Digital Microphone

Sensors

Parameter	Specification
1. **Obstacle Detection**	3 x Ultra Sonic HC-SR04 1 x Vibration sensor SW-420 1 x VL53L0X TOF Based LIDAR Laser Distance Sensor 2 x IR Sensor
2. **Navigation Support**	Web Cam
3. **Special Sensors**	1 x Vibration sensor SW-420 1 x VL53L0X TOF Based LIDAR Laser Distance Sensor

Cleaning Mechanism & Performance

Parameter	Specification
Cleaning Mechanism Type	Tornado-size suction mechanism (Cyclonic action with dual brush assist)
Storage Capacity	470 mL
Container Features	Detachable dust container, easy-to-clean, transparent body for dust visibility, and secure lock lid to prevent spillage
Max Coverage Speed	~0.25 m/s (≈ 15 m/min)
Obstacle Avoidance	Real-time detection using ultrasonic and LiDAR sensors (effective within 2–150 cm range)
Garbage Types Handled	Dust, small debris, paper scraps, hair, lightweight dry particles
Efficiency (Prototype Test)	~85% cleaning efficiency in controlled indoor environment (1 m^2 test zone)

Unique Features

Parameter	Specification
Unique Features	1.Dual Cleaning System (Brush + Vacuum) 2.Autonomous with Obstacle Avoidance 3.Modular Dustbin with Quick Release 4.Adaptive Suction for Power Saving 5.Lightweight and Cost-effective Design